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Omnidirectional slip bowl

Abstract

An omnidirectional slip bowl includes a support frame having a central axis and defining an opening therein to receive an oilfield tubular passing into or coming out of a wellbore. A plurality of pivot arms moves between an open configuration and a closed configuration. A plurality of actuating mechanisms each having a first end connected to one of the plurality of pivot arms, wherein each of the plurality of actuating mechanisms moves the associated pivot arm between the open configuration and the closed configuration. A plurality of gripping assemblies each associated with one of the plurality of pivot arms lock the tubular in a fixed position and prevent movement of the tubular from moving into or out of the wellbore. When the plurality of pivot arms are in the closed configuration and the tubular is moving out of the wellbore, the plurality of gripping assemblies move in a first direction along the pivot arm and engage the tubular to prevent moving out of the wellbore. When the plurality of pivot arms are in the closed configuration and the tubular is moving into the wellbore, the plurality of gripping assemblies move in a second direction along the pivot arm and engage the tubular to prevent moving into the wellbore.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to slip bowls for snubbing equipment and hydraulic work over units, and more particularly, to a slip bowl that can prevent oilfield tubulars used downhole from movement in both the in-hole direction and the out-of-hole direction simultaneously.

BACKGROUND

(2) A slip bowl is used to maintain control over oilfield tubulars such as drill string, production pipe, or well bore casing as they are being introduced into or removed from a well bore. For purposes of this application, unless otherwise indicated the term “tubulars” and “oilfield tubulars” refer to drill pipe, drill strings, production pipe, production strings, jointed pipe and collars, jointed and continuous tubing, casing and other types of oilfield tubular members and strings formed of such tubular members. Slip bowls have been an integral part of the snubbing service industry for the past 50 years. Existing sizes and styles of traditional slip bowls are designed to support pipe weight in only one direction. The slip bowl operates as a mechanical check valve, allowing pipe to move freely in one direction but not in the opposite direction when the bowl is closed. Current designs require snubbing units to have a set of two slip bowls. Each set has one bowl right side up to handle pipe-heavy weight (i.e., load directed down into the well bore when the pipe weight exceeds the force from well pressure) and one bowl upside down to handle pipe-light weight (i.e., load directed out of the well bore when the force from well pressure exceeds the pipe weight). Thus, two bowls are required to hold the pipe in two directions. Additionally, traditional slip bowls are not rated to handle rotational torque loads. Thus, there is a need for a new type of slip bowl that has the capability of supporting drill pipe in both axial directions and to transmit rotational loads.

SUMMARY

(3) The present invention, as disclosed and described herein, in one aspect thereof comprises an omnidirectional slip bowl including a support frame having a central axis and defining an opening therein to receive an oilfield tubular passing into or coming out of a wellbore. A plurality of pivot arms move between an open configuration and a closed configuration. A plurality of cylinder mounts slidably engage the support frame and move between the open configuration and the closed configuration. Each of the plurality of cylinder mounts are associated with one of the plurality of pivot arms. A plurality of actuating mechanisms each associated with one of the plurality of pivot arms have a first end connected to one of the plurality of pivot arms and a second end connected to one of the plurality of cylinder mounts. Each of the plurality of actuating mechanisms moves the associated pivot arm and the associated cylinder mount between the open configuration and the closed configuration. A plurality of pipe inserts engage the tubular in the closed configuration to prevent the tubular from moving perpendicularly to the central axis of the support frame. A plurality of carrier plates each associated with one of the plurality of pivot arms has a first side for supporting a pipe insert of the plurality of pipe inserts and a second side defines a plurality of alternately angled surfaces. A first portion of the plurality of alternately angled surfaces are angled in a first direction and a second portion of the plurality of alternately angled surfaces are angled in a second direction. A plurality of slide plates, each of the plurality of slide plates has a first side for engaging a corresponding surface on one of the plurality of pivot arms and a second side defining a plurality of alternately angled surfaces. A first portion of the plurality of alternately angled surfaces are angled in a first direction and a second portion of the plurality of alternately angled surfaces are angled in a second direction.

(4) When the plurality of pivot arms are in the closed configuration and the tubular is moving out of the wellbore, the slide plate moves in a first direction along the pivot arm and the first portion of the plurality of alternately angled surfaces angled in the first direction of the slide plate engage the first portion of the plurality of alternately angled surfaces angled in first direction of the carrier plate to cause the pipe insert associated with the carrier plate and the slip plate to more firmly engage the tubular to prevent moving out of the wellbore. When the plurality of pivot arms are in the closed configuration and the tubular is moving into the wellbore, the slide plate moves in a second direction along the pivot arm and the second portion of the plurality of alternately angled surfaces angled in the second direction of the slide plate engage the second portion of the plurality of alternately angled surfaces angled in the second direction of the carrier plate to cause the pipe insert associated with the carrier plate and the slip plate to more firmly engage the tubular to prevent moving into the wellbore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding, reference is now made to the following description taken in conjunction with the accompanying Drawings in which:

(2) FIG. 1 illustrates a perspective view of an omnidirectional slip bowl in a closed configuration;

(3) FIG. 2 illustrates a perspective view of an omnidirectional slip bowl in an open configuration;

(4) FIG. 3 illustrates a side view of the omnidirectional slip bowl of a pivot arm in an open position;

(5) FIG. 4 illustrates a perspective view of a pivot arm in a closed position;

(6) FIG. 5 illustrates a perspective view of the pivot arm;

(7) FIG. 6 illustrates a perspective view of the slide plate and carrier assemblies;

(8) FIG. 7 illustrates a perspective view of the carrier mount; and

(9) FIG. 8 illustrates the forces that lock a pivot arm to a locking bar responsive to tubular string weight.

DETAILED DESCRIPTION

(10) Referring now to the drawings, wherein like reference numbers are used herein to designate like elements throughout, the various views and embodiments of an omnidirectional slip bowl are illustrated and described, and other possible embodiments are described. The figures are not necessarily drawn to scale, and in some instances the drawings have been exaggerated and/or simplified in places for illustrative purposes only. One of ordinary skill in the art will appreciate the many possible applications and variations based on the following examples of possible embodiments.

(11) Referring now to the drawings, and more particularly to FIGS. **1** and **2**, there is illustrated perspective views of the omnidirectional slip bowl in both the closed configuration (FIG. **1**) and open configuration (FIG. **2**). The omnidirectional slip bowl **102** has a support frame **104** including a top plate **106** and the lower plate **108** that are interconnected by a central support frame **110**. While the current embodiment illustrates a support frame other shapes of the support frame may be used. The top plate **106**, the lower plate **108** and the central support frame **110** define a circular opening therein to enable tubulars to pass through the support frame **104**. A number of pivot arms **112** are pivotally connected to the central support frame **110** at respective pivot pins **114**. While FIGS. **1** and **2** illustrate the use of three pivot arms **112** within the slip bowl **102**, it will be appreciated by one skilled in the art that a greater or lesser number of pivot arms **112** may be used for gripping tubulars inserted within the slip bowl **102** in order to prevent the tubulars from moving within the slip bowl.

(12) The pivot arms **112** pivot between the closed position as illustrated in FIG. **1** and an open position as illustrated in FIG. **2**. Each pivot arm **112** is actuated between the closed position and the open position utilizing a pair of hydraulic cylinders **116**. The hydraulic cylinders **116** pivotally interconnect at a first end to the pivot arm **112** at a support bracket **118** and to a cylinder mount **120** at a second end of the hydraulic cylinder **116**. When in the closed position, the slip bowl **102** is capable and rated for transmitting rotational torque loads to tubulars gripped by the slip bowl.

(13) The pivot arm **112** has mounted thereon a slip plate **122**. When the slip bowl is in the closed configuration, the slide plate **122** moves up and down along an interior arc-shaped face of the pivot arm **112** in order to maintain engagement of the slip bowl with the tubular located within the slip bowl **102**. The slide plate **122** engages a carrier **124** that moves up and down along the inner face of the pivot arm **112** in association with the slide plate **122**. A gripping insert **126** is mounted within the carrier **124**. The gripping insert **126** includes a surface, as will be more particularly described hereinbelow, for engaging the surface of the pipe or casing that is inserted within the slip bowl **102**. The gripping insert **126** prevents the tubular from moving perpendicularly to the central axis of the support frame **104** when gripping the tubular.

(14) Referring now to FIG. **3**, there is illustrated a cutaway side view of a pivot arm **112** in an open position. As mentioned previously, the first end of the pivot arm **112** is pivotally mounted to the central support frame **110** on a pin **114**. The pivot arm **112** rotates between the open position illustrated in FIG. **3** to the closed position (FIG. **4**) wherein a hook member **302** located on the second end of the pivot arm **112** opposite the first end of the pivot arm engages the cylinder mount **120**. The hook member **302** defines a seat for a latching bar **304** associated with the cylinder mount **120**. When the pivot arm **112** is moved to the closed position, the latching bar **304** rests within the locking face **306** of the hook member **302**. As will be more fully described hereinbelow, when the slip bowl **102** is engaging a tubular with the pivot arms **112**, the hook member **302** remains firmly engaged with the latching bar **304** and prevents the pivot arm **112** from pivoting from the closed position to the open position.

(15) Referring now also to FIG. **4**, a retaining arm **404** extends from the back of the carrier **124**. The retaining arm **404** base inserts into a slot defined by the slide plate **122** and the pivot arm **112** and is held in place by a retaining pin **406**. The retaining arm **404** enables the carrier **124** and the slide plate **122** to be held against biasing springs within the biasing spring chambers **402**.

(16) The slip plate **122** comprises an elongated member having a first side (i.e., outer face) thereof having an arcuate shape that engages with a similar arcuate shape on an interior face of the pivot arm **112**. The curved shape of the outer face of the slide plate **122** and the corresponding curved shape of the inner face of the pivot arm **112** enable the slide plate to move up and down along the pivot arm **112**. The arcuate shape of the slip plate **122** allows the carrier **124** to self-adjust and remain parallel to the centerline of the tubular in the wellbore. If for instance, the tubular was significantly undersized or the gripping insert worn and dull, the pivot arms **112** would travel further than nominal towards the centerline of the tubular before the insert would meet the tubular. Without the circular surface the gripping insert **126** would only contact the tubular at the insert's top edge. The circular sliding surface of the slip plate **122** allows the carrier **124** to shift with respect to the pivot arm **112** and align with the surface/centerline of the tubular. As the pivot arm **112** pivots closer to the centerline of the tubular, the slip plate **122** will remain parallel to the centerline tubular and drop down closer to the pivot of the pivot arm as it slides along the circular surface. The slip bowl cylinders **116** will position the pivot arms **112** until the gripping inserts **126** contact the tubular and the pre-load springs have been fully compressed. Then as the slip bowl is loaded, the angles on the second surface of the slip plate **122** will force the pivot arm **112** out until the hook **302** rests against the locking pin **304**.

(17) A second side (i.e., inner face) of the slide plate **122** defines a series of alternating angled surfaces **308** that are angled in a first direction **308A** or a second direction **308B**. By providing the alternating angled surfaces **308**, the slide plate **122** allows movement of the carrier **124** and gripping insert **126** to enable the gripping insert to grip and hold a section of tubular in both of out-hole and in-hole directions. In this manner, the same group of pivot arms **112** and a single slip bowl **102** can prevent the tubular from moving in either direction in or out of the wellbore.

(18) Referring now to FIG. 4, there is illustrated a cross-sectional perspective view through the center of the pivot arm **112**. The cross-sectional view illustrates the pivot arm **112** in the locked position with respect to the latching bar **304** of the cylinder mount **120**. The latching bar **304** is secured to the locking face **306** of the hook member **302**. The pivot arm **112** further defines a pair of slide plate biasing chambers **402** into which preload springs are inserted for biasing the slide plate **122** outward from the pivot arm **112**.

(19) FIG. 4 also more particularly illustrates the pipe teeth **418** of the gripping inserts **126**. Each of the gripping inserts **126** have a curved inner face **420** that is similar to the surface of a tubular to be gripped by the omnidirectional slip bowl **102**. The curved outer face **420** includes a plurality of teeth therein that enable frictional engagement between the gripping insert **126** and a tubular. The gripping inserts **126** are made of material with a hardness value equal to or greater than the tubular to be gripped. The teeth **418** may also comprise a replaceable insert that may be periodically replaced as the teeth **418** wear down from repeated use. The gripping inserts **126** may comprise teeth, a grit face or even a smooth face depending on the application. The pipe inserts are selected based on the type and size of tubular being gripped by the slip bowl.

(20) FIG. 5 more particularly illustrates a perspective view of only the pivot arm **112**. As discussed previously, a first end of the pivot arm **112** defines an opening **502** for receiving a pin **114** that is inserted through the pivot arm and the central support frame **110** to enable the pivot arm **112** to pivot between the open position and a closed position. The pivot arm **112** further defines a retainer clip opening **504** through which retaining pin **406** may be inserted to secure the retaining arm **404** that secures the carrier **124** and the slide plate **124** to the pivot arm **112**. A further opening **506** is defined on each side of the pivot arm **112** for receiving a pin (i.e., through the support **118**) to secure the hydraulic cylinder **116** to the pivot arm **112** to enable movement between the open position and a closed position as described hereinabove. The second end of the pivot arm **112** includes the hook member **302** for securing to the latching bar **304** of the cylinder mount **120**. The cylinder mount engages with locking face **306** to maintain the pivot arm **112** in the closed position when a tubular is being gripped by the slip bowl **102**. The pivot arm **112** further includes a

retaining slot **508** for receiving the retaining arm **404**. The retaining arm **404** inserts through a slot in the slide plate **122** in order to maintain the slide plate against the biasing springs inserted into the biasing chambers **402**.

(21) Referring now to FIG. **6**, there is illustrated a perspective view of the slide plate **122** and the carrier **124**. The slide plate **122** has a slight arcuate shape from a first end **603** to a second end **605** thereof. An outer face **602** of the slide plate **122**, which rests against the pivot arm **112** defines a smooth arc from the first end **603** to the second end **605**. An inner face **604** of the slide plate **122** includes a series of oppositely angled surfaces **606** and **608** to create a series of peaks and valleys on the inner face. A first portion **606** of the surfaces are angled in a first direction while a second portion **608** of the surfaces are angled in a second direction. The oppositely angled surfaces enable the slide plate **122** to configure the slip bowl to engage and grip the tubular that is either being pushed down into a wellbore or pulled out of the well bore. When the slip bowl **102** is gripping a tubular within the slip bowl and the weight of the tubular is pulling the pipe down into the wellbore, the second angled faces **608** of the slip plate **122** are wedgingly engaged with adjoining faces of the carrier **124**. In this pipe-heavy situation, the greater the downward movement of the carrier **124** relative to the slide plate **122**, the greater the lateral wedging force produced between the second angled faces **608** of the slide plate **122** and the adjacent angled faces of the carrier **124**. This wedging force pushes the carrier **124** and gripping insert **126** against the tubular while simultaneously pushing the slide plate **122** and pivot arm **112** away from the tubular. Similarly, when the slip bowl **102** is gripping a tubular that is being forced upward out of the wellbore, the first angled surfaces **606** of the slide plate **122** wedgingly engage adjoining angled surfaces on the carrier **124** to prevent movement of the tubular out of the slip bowl **102**. In this pipe-light situation, the greater the upward movement of the carrier **124** relative to the slide plate **122**, the greater the lateral wedging force produced between the first angled faces **606** of the slide plate **122** and the adjacent angled faces of the carrier **124**.

(22) As further described herein, when the slip **102** is gripping the tubular in either a pipe-heavy or pipe-light configuration, the outward lateral force on the pivot arm **112** caused by the wedging action between the slide plate **122** and carrier **124** is transmitted up the pivot arm to the hook member **302**, causing the hook member to likewise exert an outward lateral force against the latching bar **304**. The greater the wedging force on the pivot bar **112**, the greater the locking force between the hook member **302** and the latching bar **304**, regardless of whether the load is from a pipe-heavy or pipe-light situation. This interlocking relationship helps ensure the latching bar **304** cannot disengage from the hook member **302** and open the slip **102** when a significant load is present.

(23) The carrier **124** includes on an outer face **609** thereof a series of oppositely angled surfaces **610** and **612**. A first portion **610** of the surfaces are angled in a first direction while a second portion **612** of the surfaces are angled in a second direction to provide a series of peaks and valleys and, when in an unloaded configuration, where the peaks of the carrier plate substantially align with the valleys of the slide plate **122** and the valleys of the carrier substantially align with the peaks of the slide plate. The oppositely angled surfaces enable the carrier **124** to configure the slip bowl **102** to engage and grip a tubular that is either being pushed down into a wellbore or pulled out of the wellbore. When the slip bowl **102** is gripping a tubular within the slip bowl and the weight of the string is pulling the pipe down into the wellbore, the first angled faces **610** of the carrier **124** are wedgingly engaged with an adjoining angled faces **608** of the slide plate **602**. Similarly, when the slip bowl **102** is gripping tubular that is being forced upward out of the wellbore, the second angled surfaces **612** of the carrier **124** wedgingly engage adjoining angled surfaces **610** on the slide plate **122** to prevent movement of the tubular out of the slip bowl. The inner surface **616** of the carrier **124** includes slots **618** for receiving the gripping inserts **126** that directly engage the tubular and a pivoting member **128** to secure a gripping insert **126** into the carrier **124**. The carrier **124** has extending perpendicularly from the back thereof a retaining arm

404 defining an opening **620** therein that is inserted through a slot **622** defined within the slide plate **122**. As described hereinabove, when the retaining arm **404** is inserted through the slot **622** of the slide plate **602** and further through a corresponding slot **508** within the pivot arm **112**, a pin may be used to insert through the pivot arm and the opening **620** to retain the carrier **124** and slide plate **122** against the pivot arm **112**.

(24) In some embodiments, the wedging surfaces on the slide plate **122** and carrier **124** each comprise at least one set of first and second flat surfaces, where each respective first flat surface is angled at a constant positive slope and each respective second flat surface is angled at a constant negative slope (with the slopes being measured relative to a median line on the respective wedging surface). In some such embodiments, multiple sets of first and second flat surfaces are arranged consecutively along the wedging surfaces (also known as “sawtooth” profile). For example, in the embodiment of FIG. 6, the wedging surfaces on the slide plate **122** and carrier **124** each comprise at least four adjacent sets of first and second flat surfaces arranged consecutively, where each respective first flat surface is angled at a constant positive slope and each respective second flat surface is angled at a constant negative slope. In other embodiments, the wedging surfaces on the slide plate **122** and carrier **124** each comprise a continuously curving surface having at least a first curved region of positive slope and a second curved region of negative slope. In some such embodiments, the wedging surfaces on the slide plate **122** and carrier **124** each comprise a continuously curving surface having multiple alternating portions of positive slope and negative slope. In some such embodiments, the wedging surfaces on the slide plate **122** and carrier **124** each comprise a repeating sine curve having multiple alternating portions of positive slope and negative slope. In still further embodiments, the wedging surfaces on the slip plate **122** and carrier **124** comprise other profiles that, when placed in contact with one another and moved relative to one another along a travel line, cause wedging engagement with one another producing a lateral force between the wedging surfaces, wherein the direction of the lateral force is the same when the direction of movement along the travel line is positive or negative.

(25) Referring now to FIG. 7, there is illustrated the cylinder mount **120**. The cylinder mount **120** includes a pair of hydraulic cylinders **116**. The cylinders include a first end **702** defining a piston arm connector **704** that interconnect with the pivot arms **112**. The second end **706** of the hydraulic cylinder **116** connects with the cylinder mount **120** at a flange **710** that inserts into a U-shaped connector **712** that extends downward from the cylinder mount **120**. The cylinder mount **120** comprises a substantially rectangularly shaped member **708** from which the U-shaped connectors hang downward to engage the hydraulic cylinders **116**. A pin is inserted through the U-shaped connector **712** and the flange **710** in order to interconnect the hydraulic cylinder **116** to the cylinder mount **120**. The locking bar **304** extends downward from the cylinder mount **120** from a pair of cylindrical members **716**. The cylindrical members **716** extend downward from the cylinder mount **120** to maintain the locking bar **304** a fixed distance below the rectangular structure of the cylinder mount **120**. The cylindrical members **716** are spring biased using nitrogen gas springs in one embodiment but other biasing mechanisms may be used. The locking bar **304** when engaged by the hook **306** of the pivot arm **112** maintains the pivot arms in a locked position when the slip bowl is bearing a string weight as more particularly described in FIG. 8.

(26) When the pivot arms **112** are in a closed position and the slip bowl **102** is supporting a tubular from either falling into or pushing out of a wellbore, the cylinder mount **120** locks the pivot arms in the closed position. This is more particularly illustrated in FIG. 8. When a pivot arm **112** is in a closed position and the tubular (not shown) is engaged by the gripping inserts **126**, the forces within the omnidirectional slip bowl **102** are as illustrated in FIG. 8.

(27) Once there is tubular string weight held by the slip bowl **102**, the tubular weight may in one embodiment provide a downward force in the direction shown by arrow **802**. The downward force **802** of the tubular weight forces the carriers **124** to move downward relative to the slide plates **122** causing the adjoining angled surfaces to wedge against one another and produce an outward lateral

force on the slide plates **122** shown by arrow **804**. This causes the slide plate **122** to provide a lateral force on the pivot arms **112** in the direction illustrated by arrows **804**. The lateral force **804** upon the pivot arm **112** also causes a lateral force between the hook **302** at the top of the pivot arm **112** and the latching bar **304** as illustrated generally by the arrows **806**. The lateral force causes frictional forces between the hook **302** and latching bar **304** to help lock the latching bar in place and keep the cylinders **116** from being able to unlatch the pivot arms **112** when tubular weight is present in either direction.

(28) It will be appreciated by those skilled in the art having the benefit of this disclosure that this omnidirectional slip bowl provides a single slip bowl for supporting tubular weight either down into or out of the drill hole. It should be understood that the drawings and detailed description herein are to be regarded in an illustrative rather than a restrictive manner, and are not intended to be limiting to the particular forms and examples disclosed. On the contrary, included are any further modifications, changes, rearrangements, substitutions, alternatives, design choices, and embodiments apparent to those of ordinary skill in the art, without departing from the spirit and scope hereof, as defined by the following claims. Thus, it is intended that the following claims be interpreted to embrace all such further modifications, changes, rearrangements, substitutions, alternatives, design choices, and embodiments.

Claims

1. An omnidirectional slip bowl comprising: a support frame having a central axis and defining an opening therein to receive a tubular passing into or coming out of a wellbore; a plurality of pivot arms for moving between an open configuration and a closed configuration; a plurality of cylinder mounts slidably engaging the support frame and moving between the open configuration and the closed configuration, each of the plurality of cylinder mounts associated with one of the plurality of pivot arms; a plurality of actuators each having a first end connected to one of the plurality of pivot arms and a second end connected to one of the plurality of cylinder mounts, wherein each of the plurality of actuators moves the associated pivot arm and the associated cylinder mount between the open configuration and the closed configuration; a plurality of pipe inserts for engaging the tubular in the closed configuration to prevent the tubular from moving perpendicular to the central axis of the support frame; a plurality of carriers each associated with one of the plurality of pivot arms having a first side for supporting a pipe insert of the plurality of pipe inserts and a second side defining a first plurality of alternately angled surfaces, wherein a first portion of the first plurality of alternately angled surfaces are angled in a first direction and a second portion of the first plurality of alternately angled surfaces are angled in a second direction; a plurality of slide plates, each of the plurality of slide plates having a first side for engaging a corresponding surface on one of the plurality of pivot arms and a second side defining a second plurality of alternately angled surfaces, wherein a first portion of the second plurality of alternately angled surfaces are angled in the first direction and a second portion of the second plurality of alternately angled surfaces are angled in the second direction; wherein when the plurality of pivot arms are in the closed configuration and the tubular is moving out of the wellbore, the plurality of slide plates move in a third direction along the plurality of pivot arms and the first portion of the second plurality of alternately angled surfaces angled in the first direction of the plurality of slide plates engage the first portion of the second plurality of alternately angled surfaces angled in the first direction of the plurality of carriers to cause the plurality of pipe inserts associated with the plurality of carriers and the plurality of slide plates to more firmly engage the tubular to prevent the tubular moving out of the wellbore; and wherein when the plurality of pivot arms are in the closed configuration and the tubular is moving into the wellbore, the plurality of slide plates move in a fourth direction along the plurality of pivot arms and the second portion of the second plurality of alternately angled surfaces angled in the second direction of the plurality of slide plates engage the second portion of the

second plurality of alternately angled surfaces angled in the second direction of the plurality of carriers to cause the plurality of pipe inserts associated with the plurality of carriers and the plurality of slide plates to more firmly engage the tubular to prevent the tubular moving into the wellbore.

2. The omnidirectional slip bowl of claim 1, wherein the first side of the plurality of slide plates further comprise an arcuate surface enabling the plurality of slide plates to move between a first position when the tubular is moving out of the wellbore and a second position when the tubular is moving into the wellbore.

3. The omnidirectional slip bowl of claim 2, wherein the arcuate surface of the plurality of slide plates enables the plurality of slide plates to move along the plurality of pivot arms to enable the plurality of carriers engaging the plurality of slide plates to self-adjust and remain parallel to a centerline of the tubular.

4. The omnidirectional slip bowl of claim 1, wherein each of the plurality of cylinder mounts further comprises: a member for slidably engaging a top surface of the support frame in a direction parallel to the central axis of the support frame; a u-shaped connector located at each end of the member and integrally connected therewith for connecting the plurality of cylinder mounts with the plurality of actuators; and a locking bar connected to the member and extending downward therefrom, wherein the locking bar engages a pivot arm of the plurality of pivot arms in the closed configuration to maintain the pivot arm in the closed configuration.

5. The omnidirectional slip bowl of claim 4, wherein the pivot arm maintains a locking connection with the locking bar by increasing forces between the pivot arm and the locking bar responsive to tubular weight forces applied to the omnidirectional slip bowl.

6. The omnidirectional slip bowl of claim 1, wherein the plurality of carriers further comprises: a retaining member extending from the second side of the plurality of carriers into a first opening defined within the plurality of slide plates and a second opening defined within the plurality of pivot arms; and a retaining pin for insertion through the plurality of pivot arms and the retaining member to secure the plurality of carriers and the plurality of slide plates to the plurality of pivot arms.

7. The omnidirectional slip bowl of claim 1 further comprising: wherein the plurality of pivot arms further defines a plurality of bias spring chambers therein; and a bias spring inserted into each of the bias spring chambers, wherein the bias spring biases the second side of the plurality of slide plates to engage the second side of the plurality of carriers.

8. The omnidirectional slip bowl of claim 1, wherein when each of the plurality of pivot arms are in the closed configuration, the omnidirectional slip bowl transmits rotation torque loads to the tubular.

9. An omnidirectional slip bowl comprising: a support frame having a central axis and defining an opening therein to receive a tubular passing into or coming out of a wellbore; a plurality of pivot arms for moving between an open configuration and a closed configuration; a plurality of actuators each having a first end connected to one of the plurality of pivot arms, wherein each of the plurality of actuators moves the associated pivot arm between the open configuration and the closed configuration; a plurality of gripper assemblies, each gripper assembly associated with one of the plurality of pivot arms for locking the tubular in a fixed position and prevent movement of the tubular from moving into or out of the wellbore; wherein when the plurality of pivot arms are in the closed configuration and the tubular is moving out of the wellbore, the plurality of gripper assemblies move in a first direction along the plurality of pivot arms and engage the tubular to prevent moving out of the wellbore; and wherein when the plurality of pivot arms are in the closed configuration and the tubular is moving into the wellbore, the plurality of gripper assemblies move in a second direction along the pivot arm and engage the tubular to prevent moving into the wellbore.

10. The omnidirectional slip bowl of claim 9, wherein each of the plurality of gripper assemblies

further comprises: a pipe insert for engaging the tubular in the closed configuration to prevent the tubular from moving perpendicular to the central axis of the support frame; a carrier associated with one of the plurality of pivot arms has a first side for supporting the pipe insert and a second side defining a first plurality of alternately angled surfaces, wherein a first portion of the first plurality of alternately angled surfaces are angled in a first direction and a second portion of the first plurality of alternately angled surfaces are angled in a second direction; and a slide plate having a first side for engaging a corresponding surface on one of the plurality of pivot arms and a second side defining a second plurality of alternately angled surfaces, wherein a first portion of the second plurality of alternately angled surfaces are angled in the first direction and a second portion of the second plurality of alternately angled surfaces are angled in the second direction.

11. The omnidirectional slip bowl of claim 10, wherein the first side of the slide plate comprises an arcuate surface enabling the slide plate to move between a first position when the tubular is moving out of the wellbore and a second position when the tubular is moving into the wellbore.

12. The omnidirectional slip bowl of claim 11, wherein the arcuate surface of the slide plate enables the slide plate to move along the pivot arm to enable the carrier engaging the slide plate to self-adjust and remain parallel to a centerline of the tubular.

13. The omnidirectional slip bowl of claim 10 further comprising: wherein when the slide plate moves in a first direction along the pivot arm and the first portion of the second plurality of alternately angled surfaces angled in the first direction of the slide plate engage the first portion of the plurality of alternately angled surfaces angled in first direction of the carrier to cause the pipe insert associated with the carrier and the slide plate to more firmly engage the tubular to prevent moving out of the wellbore; and wherein when the slide plate moves in a second direction along the pivot arm and the second portion of the second plurality of alternately angled surfaces angled in the second direction of the slide plate engage the second portion of the plurality of alternately angled surfaces angled in the second direction of the carrier to cause the pipe insert associated with the carrier and the slide plate to more firmly engage the tubular to prevent moving into the wellbore.

14. The omnidirectional slip bowl of claim 10 further comprising: a plurality of cylinder mounts slidably engaging the support frame and moving between the open configuration and the closed configuration, each of the plurality of cylinder mounts associated with one of the plurality of pivot arms, wherein each of the plurality of cylinder mounts further comprises: a member for slidably engaging a top surface of the support frame in a direction parallel to the central axis of the support frame; a u-shaped connector located at each end of the member and integrally connected therewith for connecting the plurality of cylinder mounts with the plurality of actuators; and a locking bar connected to the member and extending downward therefrom, wherein the locking bar engages a pivot arm in the closed configuration to maintain the pivot arm in the closed configuration.

15. The omnidirectional slip bowl of claim 14, wherein the pivot arm maintains a locking connection with the locking bar by increasing forces between the pivot arm and the locking bar responsive to tubular weight forces applied to the omnidirectional slip bowl.

16. The omnidirectional slip bowl of claim 10, wherein the carrier further comprises: a retaining member extending from the second side of the carrier into a first opening defined within the slide plate and a second opening defined within the pivot arm; and a retaining pin for insertion through the pivot arm and the retaining member to secure the carrier and the slid plate to the pivot arm.

17. The omnidirectional slip bowl of claim 10 further comprising: wherein the pivot arm further defines a plurality of bias spring chambers therein; and a bias spring inserted into each of the bias spring chambers, wherein the bias spring biases the second side of the slide plate to engage the second side of the carrier.

18. An omnidirectional slip bowl comprising: a support frame having a central axis and defining an opening therein to receive a tubular passing into or coming out of a wellbore; a plurality of pivot arms for moving between an open configuration and a closed configuration; a plurality of actuators each having a first end connected to one of the plurality of pivot arms, wherein each of the plurality

of actuators moves the associated pivot arm between the open configuration and the closed configuration; a plurality of pipe inserts each associated with one of the plurality of pivot arms for engaging the tubular in the closed configuration to prevent the tubular from moving perpendicular to the central axis of the support frame; a plurality of carriers each associated with one of the plurality of pivot arms having a first side for supporting one of the plurality of pipe inserts and a second side defining a plurality of alternately angled surfaces, wherein a first portion of the plurality of alternately angled surfaces are angled in a first direction and a second portion of the plurality of alternately angled surfaces are angled in a second direction; and a plurality of slide plates each having an arcuate side for engaging a corresponding surface on one of the plurality of pivot arms and a second side defining a plurality of alternately angled surfaces, wherein a first portion of the plurality of alternately angled surfaces are angled in a first direction and a second portion of the plurality of alternately angled surfaces are angled in a second direction, wherein the arcuate side of the plurality of slide plates enables the plurality of slide plates to move along the plurality of pivot arms to enable the plurality of carriers engaging the plurality of slide plates to self-adjust and remain parallel to a centerline of the tubular; wherein when the plurality of slide plates move in a first direction along the pivot arm and the first portion of the plurality of alternately angled surfaces angled in the first direction of the plurality of slide plates engage the first portion of the plurality of alternately angled surfaces angled in first direction of the plurality of carriers to cause the plurality of pipe inserts associated with the plurality of carriers and the plurality of slide plates to more firmly engage the tubular to prevent moving out of the wellbore; wherein when the plurality of slide plates move in a second direction along the plurality of pivot arms and the second portion of the plurality of alternately angled surfaces angled in the second direction of the plurality of slide plates engage the second portion of the plurality of alternately angled surfaces angled in the second direction of the plurality of carriers to cause the plurality of pipe inserts associated with the plurality of carriers and the plurality of slide plates to more firmly engage the tubular to prevent moving into the wellbore; a plurality of cylinder mounts slidably engaging the support frame and moving between the open configuration and the closed configuration, each of the plurality of cylinder mounts associated with one of the plurality of pivot arms, wherein each of the plurality of cylinder mounts further comprises: a member for slidably engaging a top surface of the support frame in a direction parallel to the central axis of the support frame; a u-shaped connector located at each end of the member and integrally connected therewith for connecting the plurality of cylinder mounts with the plurality of actuators; and a locking bar connected to the member and extending downward therefrom, wherein the locking bar engages a pivot arm in the closed configuration to maintain the pivot arm in the closed configuration.

19. The omnidirectional slip bowl of claim 18, wherein the first side of the plurality of slide plates comprises an arcuate surface enabling the plurality of slide plates to move between a first position when the tubular is moving out of the wellbore and a second position when the tubular is moving into the wellbore.

20. The omnidirectional slip bowl of claim 18, wherein the pivot arm maintains a locking connection with the locking bar by increasing forces between the pivot arm and the locking bar responsive to tubular weight forces applied to the omnidirectional slip bowl.
